

Cyber Ontology Foundry

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Outline

- Ontology Foundries
- Cyber Ontology Landscape
- Cyber Ontology Foundry

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- **Ontology Foundries**
- Cyber Ontology Landscape
- Cyber Ontology Foundry

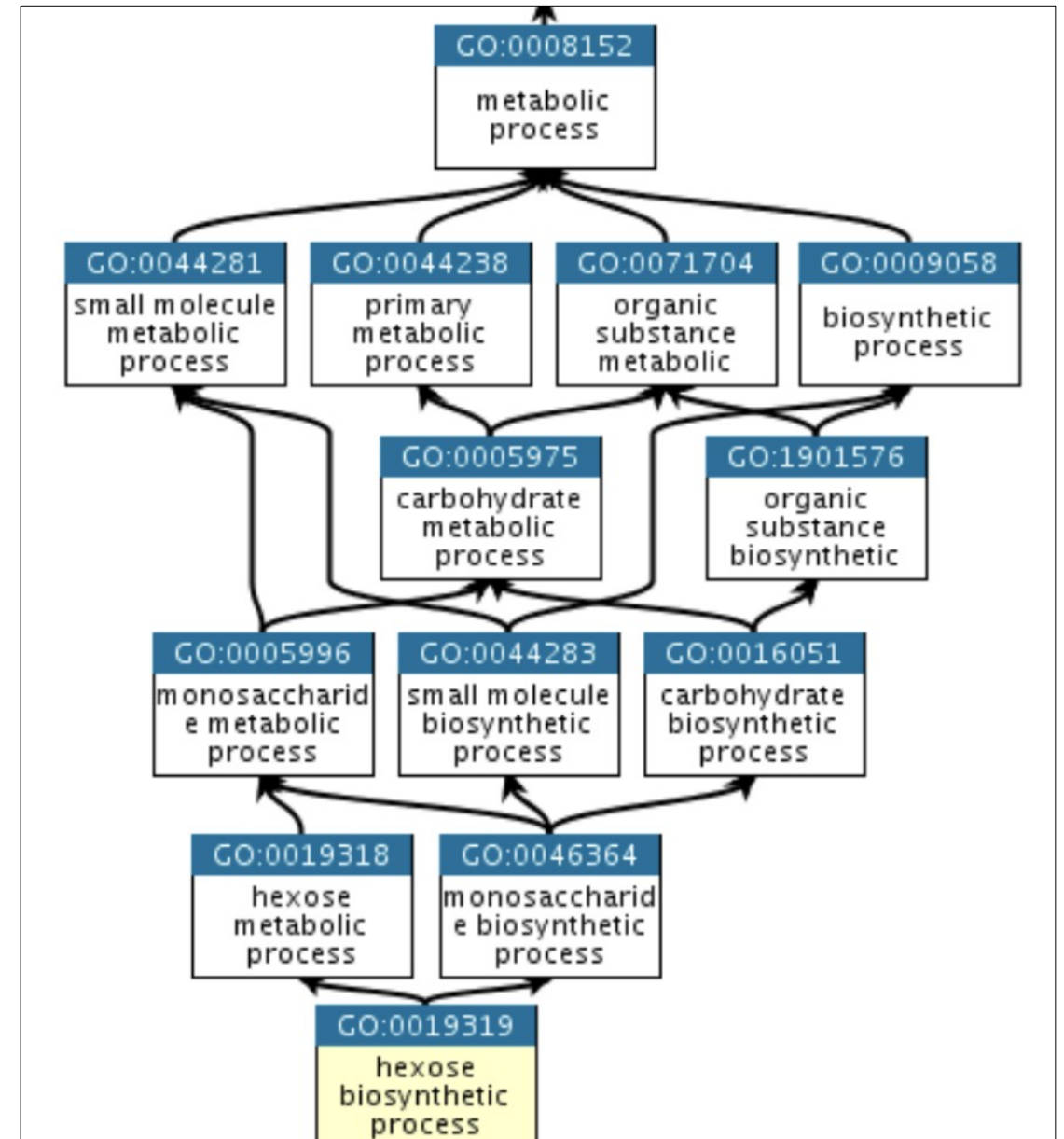
Promise of Ontology Engineering

- Ontologies are formally well-defined machine-interpretable controlled vocabularies designed to represent entities and logical relationships among them
- Ontologies make **explicit** the **implicit** meanings buried in datasets, by using basic principles of formal logic
- Ontologies provide a **semantic layer** to connect information silos

Gene Ontology - 1998



The mission of the GO Consortium is to develop a comprehensive, **computational model of biological systems**, ranging from the molecular to the organism level, across the multiplicity of species in



Proliferation of Ontologies

- When developed correctly, ontologies provide **common vocabularies** with **common semantics** across **multiple domains**
- The success of the Gene Ontology led to a proliferation of ontologies developed by subject-matter experts, computer scientists, and logicians

Proliferation of Ontologies

- When developed correctly, ontologies provide **common vocabularies** with **common semantics** across **multiple domains**
- The success of the Gene Ontology led to a proliferation of ontologies developed by subject-matter experts, computer scientists, and logicians
- Almost **none** of which were developed in coordination
- The result was **massive incompatibility** of terms and relations, confusion, in-fighting, name-calling, etc.

Open Biological and Biomedical Ontologies

- In 2005, a consortium of biologists decided to create standards for ontology development
- Such as requiring ontologies be open-source have documentation include definitions for vocabulary terms...



Overview

Open (principle 1)

Common Format (principle 2)

URI/Identifier Space (principle 3)

Versioning (principle 4)

Scope (principle 5)

Textual Definitions (principle 6)

Relations (principle 7)

Documentation (principle 8)

Documented Plurality of Users

(principle 9)

Commitment To Collaboration

(principle 10)

Locus of Authority (principle 11)

Naming Conventions (principle 12)

Notification of Changes (principle 13)

Maintenance (principle 16)

Responsiveness (principle 20)

Complexity of Industrial Manufacturing

- Industrial manufacturing creates massive, heterogeneous data
- CAD, PLM, ERP, MES, supply chain, sensor, and maintenance systems all model the world differently
- Local data models solved local problems but local success produced **global fragmentation**

Industrial Ontologies Foundry

- In 2016, the Systems Integration Division at the National Institute of Standards and Technology (NIST) brought together industrial manufacturing researchers and ontologists
- Who aimed to promote best practices for ontology development to serve interoperability.



- 1 Introduction
- 2 IOF Ontology Architecture
- 3 Common Representations
- 4 Open Availability and Copyright
- 5 Scope and Context
- 6 Modularity
- 7 Ontological Considerations
- 8 Foundational Ontology – Use
- 9 Documentation
- 10 Textual Definitions
- 11 Identifiers and Naming Conventions
- 12 IRI and Identifier Space
- 13 Versioning
- 14 Maintenance
- 15 Contributed Ontologies: – Use
- 16 Collaboration

50+ Organizations

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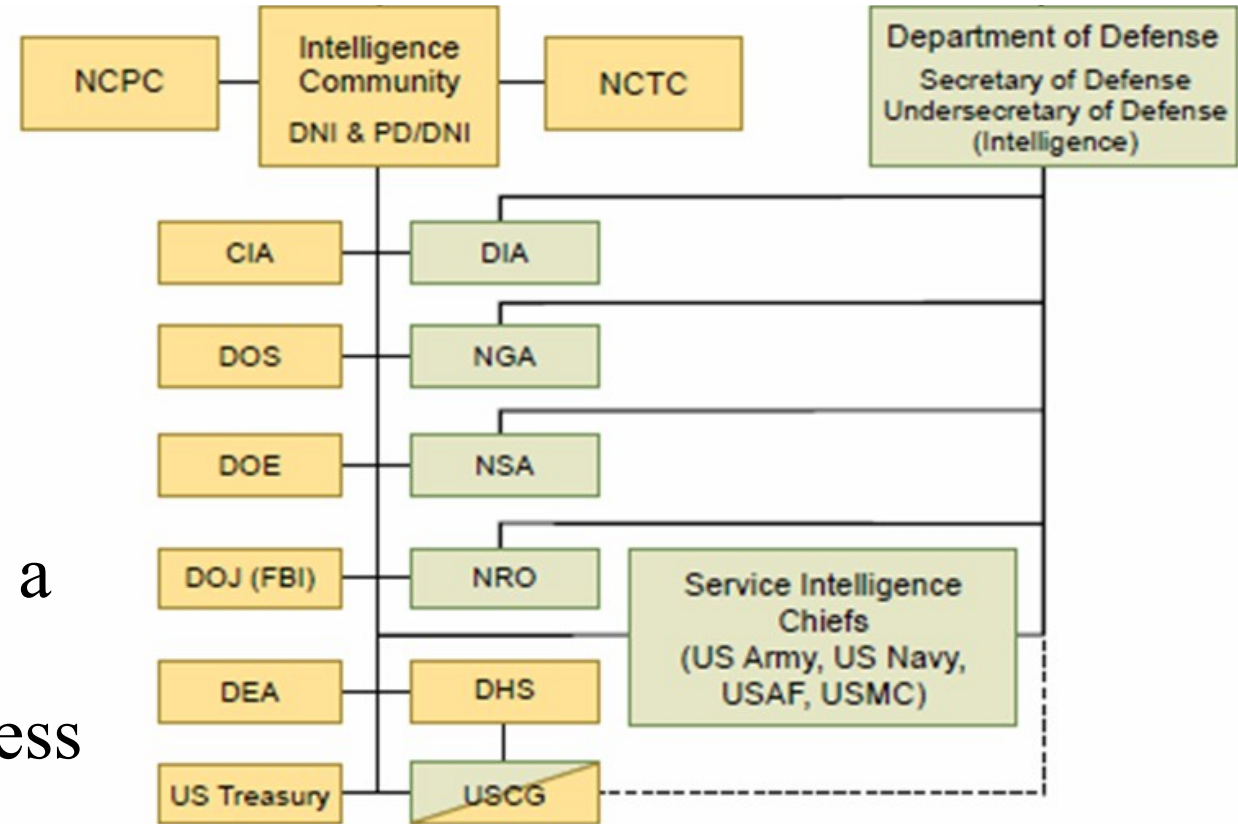


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DE VALÈNCIA



Three-Letter Suite Problem

- Various agencies create vocabularies of the terms commonly used in its domain; various ontologies were created
- Just as the Gene Ontology spurred a proliferation of uncoordinated ontologies, we should expect success to bring similar fruits
- Every agency wants to have its own “three-letter ontology suite”



National Security Ontology Foundry

- Borne out of long-standing efforts to promote interoperability among ontologies created within and across the US government.
- Aimed at constructing vetted ontology artifacts aligned with best community practices, tailored to specific US government use cases and needs.

P1 – OPENNESS	
<i>Summary</i>	
<i>Purpose</i>	
<i>Recommendations and Requirements</i>	
<i>Examples</i>	
P2 – MODIFICATION	
<i>Summary</i>	
<i>Purpose</i>	
P3 – COMMON FORMAT	
<i>Summary</i>	
<i>Recommendations and Requirements</i>	
<i>Implementation</i>	
P4 – IRI/IDENTIFIER SPACE	
<i>Summary</i>	
<i>Purpose</i>	
P5 – VERSIONING	
<i>Summary</i>	
<i>Purpose</i>	
<i>Recommendations and Requirements</i>	
<i>Implementation</i>	
P6 – SCOPE	
<i>Summary</i>	
<i>Purpose</i>	
<i>Recommendations and Requirements</i>	
<i>Example</i>	

Ontology Foundries

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**Open Biological and
Biomedical Foundry**

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Industrial Ontologies Foundry

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Summary.....

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Summary.....

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Implementation.....

P6 – SCOPE

Summary.....

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Example.....

P7 – TEXTUAL DEFINITIONS

Summary.....

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Recommendations and Requirements.....

P8 – RELATIONS

Summary.....

Purpose.....

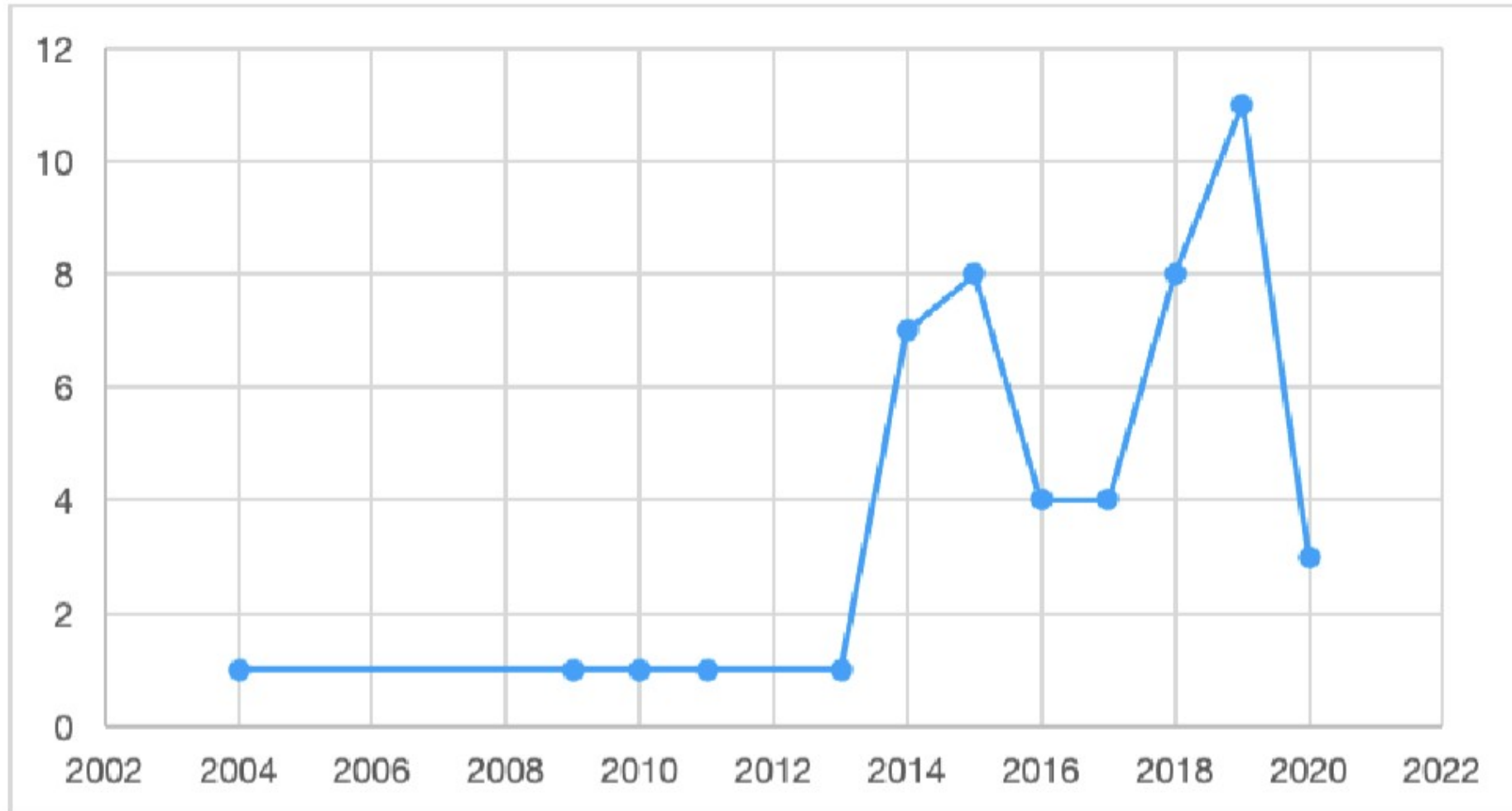
P9 – DOCUMENTATION

**National Security
Ontology Foundry**

Outline

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Figure 1. Chronology of ontologies reported in the context of cybersecurity.



Name	Scope
Unified Cyber Ontology (UCO)	General foundation for standardized representation across the cybersecurity ecosystem
Cyber-investigation Analysis Standard Expression	Ontology-based standard for cyber-investigation and digital-evidence data; built on UCO.
D3FEND	Defensive cybersecurity ontology / knowledge graph for countermeasures, artifacts, and defensive techniques
ATT&CK	Offensive cybersecurity ontology / knowledge graph for artifacts, processes, and attack techniques.
CCO Cyber Ontology	Cyber ontology extending CCO, focused on cyber assets, events, and cross-domain situational awareness.
Unified Ontology for Cyber Security	Research ontology integrating sources such as ATT&CK, D3FEND, ENGAGE, CWE, and CVE for cyber threat analysis.
Cyber security knowledge graph ontology	Research ontology developed for a cybersecurity knowledge graph, intended to integrate structured and unstructured cyber data sources.
MALOnt	Representations of malware, threat actors, and their tactics, Indicators of Compromise (IoC), and vulnerabilities

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D3FEND™

- Knowledge graph of cybersecurity countermeasure techniques
- Project aims to **standardize defensive cybersecurity technology vocabulary** in a semantic graph representing computer system components and relations to defensive and offensive cyber techniques

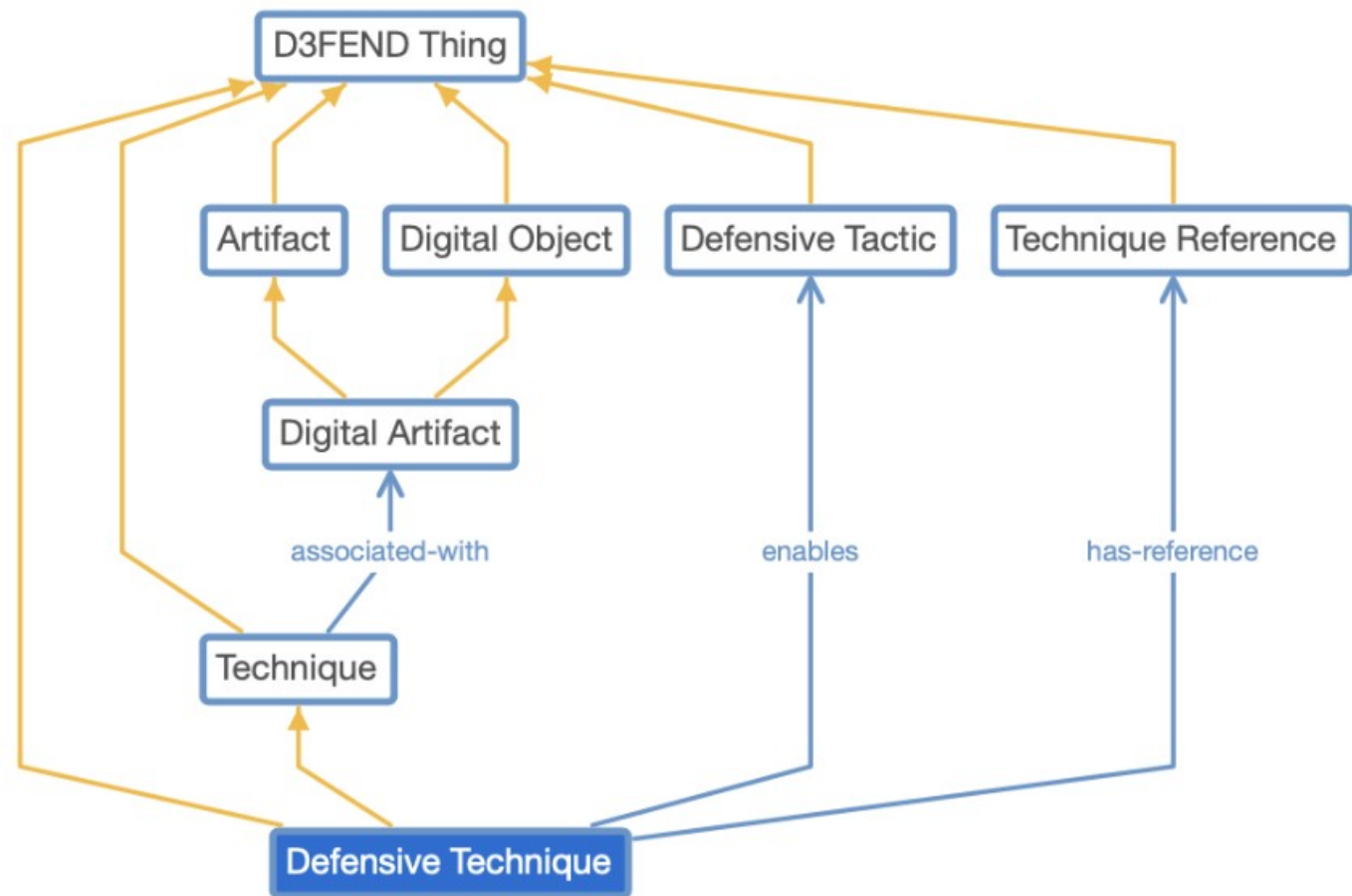


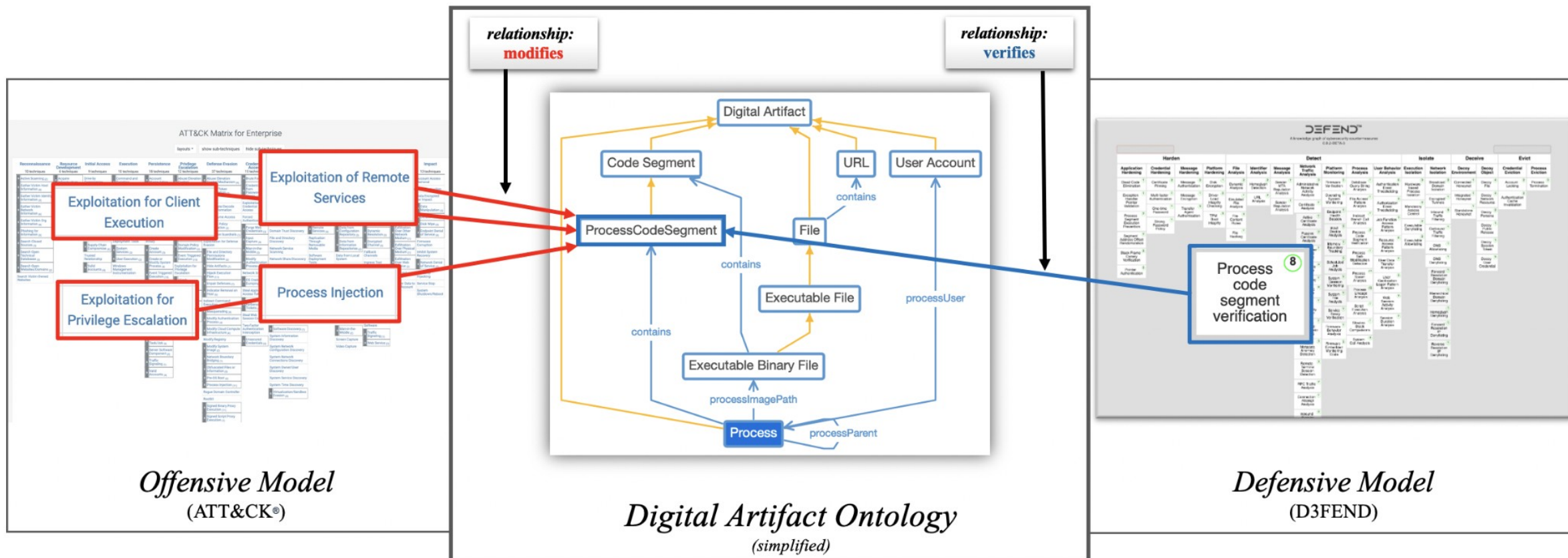
<https://github.com/d3fend>

Foundations

- The D3FEND knowledge graph **defines** key terms and relations concerning cybersecurity countermeasures
- Definitions are based on **over 500** countermeasure patents from the U.S. Patent Office between 2001-2018
- Graph supports **queries** connecting cybersecurity countermeasures to **offensive tactics, techniques, and procedures**

Metrics		
Axiom		22394
Logical axiom count		7543
Declaration axioms count		4155
Class count		2986
Object property count		211
Data property count		41
Individual count		886
Annotation Property count		37
Class axioms		
SubClassOf		4480
EquivalentClasses		0
DisjointClasses		9
GCI count		0
Hidden GCI Count		0
Object property axioms		
SubObjectPropertyOf		257
EquivalentObjectProperties		0
InverseObjectProperties		41
DisjointObjectProperties		0
FunctionalObjectProperty		0
InverseFunctionalObjectProperty		0
TransitiveObjectProperty		5
SymmetricObjectProperty		0
AsymmetricObjectProperty		0
ReflexiveObjectProperty		0
IrrefexiveObjectProperty		0
ObjectPropertyDomain		10
ObjectPropertyRange		17
SubPropertyChainOf		1





Toward a Knowledge Graph of Cybersecurity Countermeasures

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Name	Scope
STIX	Represents cyber threat intelligence: indicators, malware, campaigns, threat actors, attack patterns, incidents, relationships, etc.
TAXII	Exchanges STIX threat intelligence between systems over network services.
CACAO	Represents cyber defense playbooks and courses of action for incident response and automation.
OpenC2	Provides standardized commands for cyber-defense actions across tools and devices.
CVE	Gives common identifiers to publicly known cybersecurity vulnerabilities.
CVSS	Scores vulnerability severity using standardized metrics.
CWE	Classifies common software and hardware weakness types that can lead to vulnerabilities.
CSAF	Provides machine-readable security advisories, often for vendors publishing vulnerability information.

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STIX 2.1 — A Useful but Non-Ontological Exchange Standard

```
{
  "type": "bundle",
  "id": "bundle--a3c7e...",
  "objects": [
    {
      "type": "threat-actor",
      "spec_version": "2.1",
      "id": "threat-actor--b695c...",
      "name": "APT28",
      "threat_actor_types": ["nation-state"]
    },
    {
      "type": "attack-pattern",
      "spec_version": "2.1",
      "id": "attack-pattern--d4e2...",
      "name": "Spear Phishing"
    }
  ]
}
```

Wire format — semantics live in prose, not the model

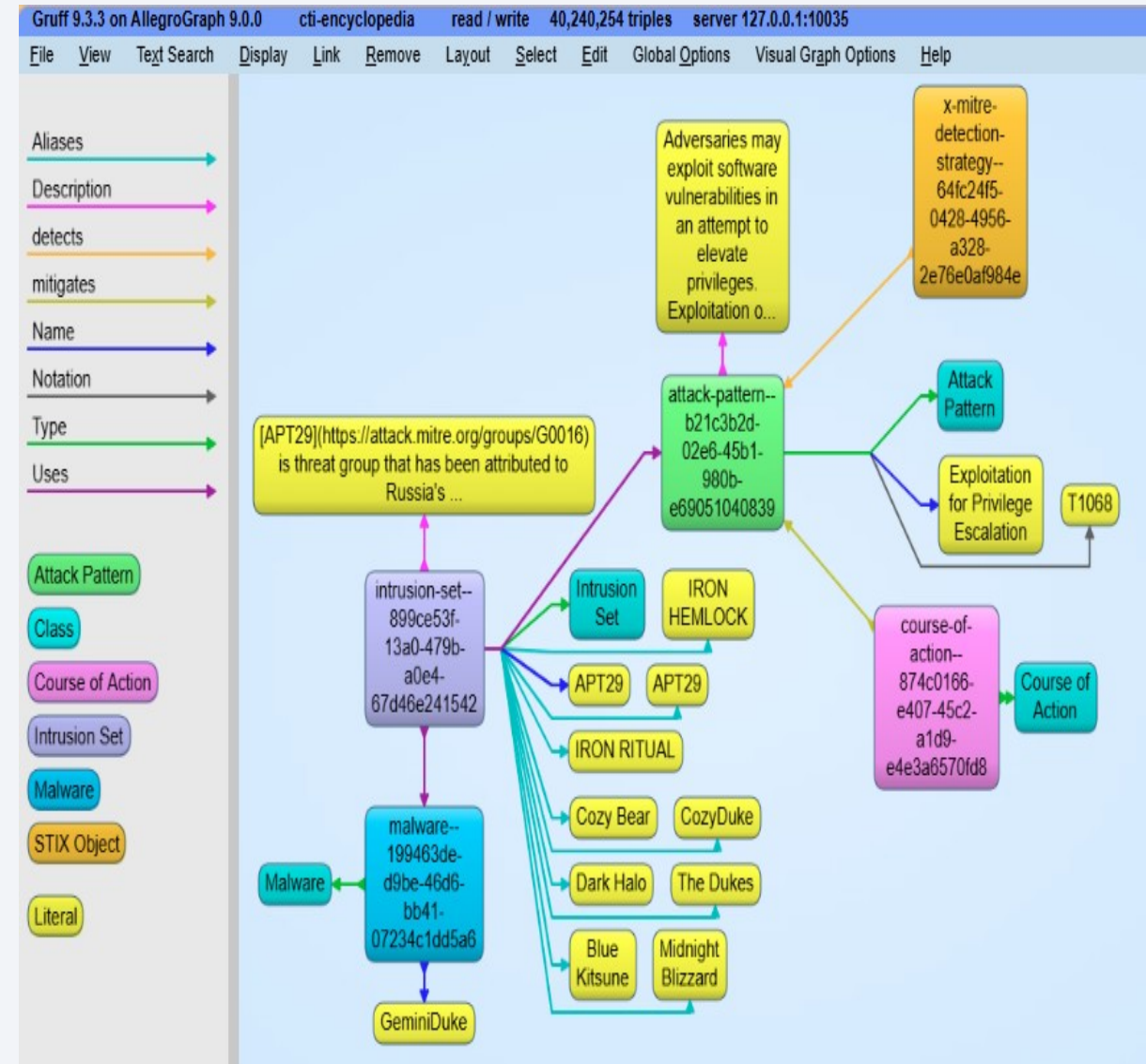


STIX 2.1 visual representation — objects and relationships

- STIX 2.1 is the OASIS standard for representing and sharing cyber threat intelligence
- Defined as a JSON schema — a data exchange format, not a knowledge representation
- Rich domain semantics exist in the specification prose, not in a formal model
- No class hierarchy, no property axioms, no entailment regime
- Cannot participate in OWL reasoning or SPARQL-based integration as-is

STONES — A Faithful Ontological Binding of STIX 2.1

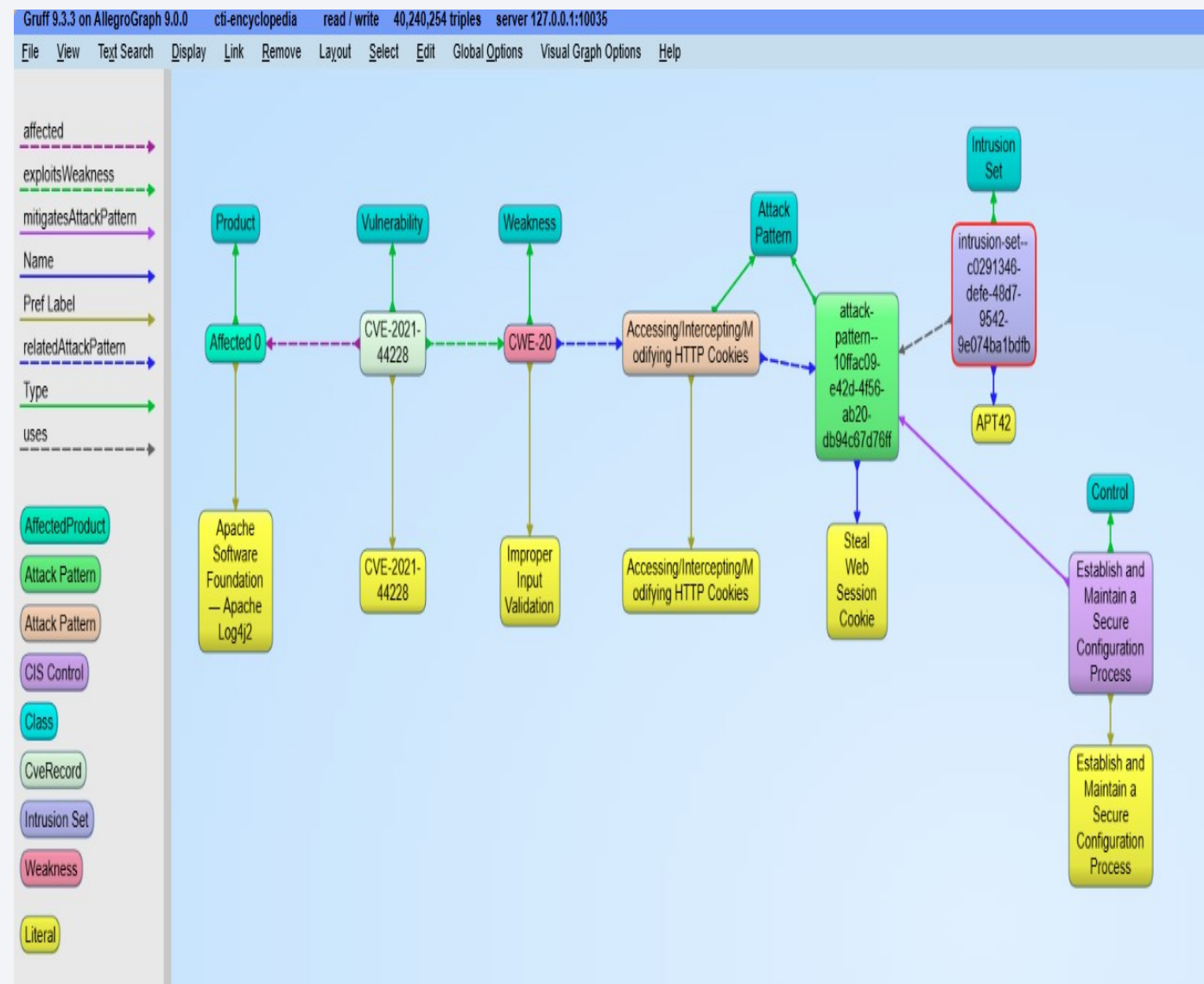
- OWL 2 ontology providing a formal representation of the STIX 2.1 specification
- Faithful by design — every STIX domain object, observable, and relationship type represented as a named class or property; semantics preserved, not reinterpreted
- Enables SPARQL querying, OWL reasoning, and knowledge graph integration over CTI data
- MIT License — independent work, not affiliated with OASIS or the CTI-TC
- cyberterrain.org/ns/stones



STONEWORK — An Extension of STONES

- OWL 2 extension ontology — imports STONES, adds what STIX 2.1 does not define
- Extends the cyber terrain to include adversary techniques (aligned with MITRE ATT&CK and D3FEND), software weaknesses (CWE), and defensive controls (NIST SP 800-53, CIS Controls)
- STONES + STONEWORK form a composable ontological stack for AI-driven CTI analysis
- cyberterrain.org/ns/stonework

Log4Shell (CVE-2021-44228) — from affected product to threat actor to defensive control, in a single SPARQL query.



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- Cyber Ontology Foundry

The Cyber Ontology Foundry

- A community-governed initiative to develop, maintain, and coordinate a suite of open, interoperable, formally grounded ontologies for the cyber domain
- Provide ontologies that clarify entities, processes, roles, artifacts, vulnerabilities, threats, controls, observations, evidence, and claims represented in and across existing cyber standards such as STIX/TAXII, D3FEND, ATT&CK, CAPEC, CVE, and so on

COF Principles

- Modeled off OBO, IOF, and NSOF principles
- With specific emphasis on **rapid development and testing**
- And leveraging ontologies as a **transparent foundation for code**

Principle	Meaning
Open	Public Foundry ontologies should be available for reuse, review, and extension.
Formalized	Ontology content should be represented in standard machine-readable formats, with OWL 2 as the minimum formal target for official modules unless an exception is approved.
Scoped	Every ontology should state what it covers, what it does not cover, and what neighboring ontologies it depends on.
Orthogonal	The Foundry should avoid unnecessary duplication and reuse existing ontology elements where possible.
Identified	Ontologies, classes, relations, and reference patterns should have stable identifiers and managed prefixes.
Defined	Classes and relations should have clear textual definitions.
Validated	Ontologies should be checked for consistency, satisfiability, and structural integrity.
Justified	Assertions, mappings, definitions, and design decisions should carry provenance and justification.
Governed	Every official ontology should have named maintainers, review procedures, and decision rights.
Reviewed	Development should be collaborative, issue-driven, and transparent.
Versioned	Releases should be versioned, archived, and accompanied by release notes.
Engaged	Maintainers should respond to community issues and evolving cyber practice.

Mapping (Surprise)

Foundry ontologies **must** be accompanied by mappings into existing Foundry ontologies, demonstrating where respective classes and relations logically overlap or differ

Table of Laplace Transforms

$f(t)$	$F(s)$	$f(t)$	$F(s)$
$\delta(t)$	1	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
1	$1/s$	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
t	$1/s^2$	$\cosh at$	$\frac{s}{s^2 - a^2}$
t^2	$2/s^3$	$\sinh at$	$\frac{a}{s^2 - a^2}$
t^n	$\frac{n!}{s^{n+1}}$	$e^{at} \cos \omega t$	$\frac{s-a}{(s-a)^2 + \omega^2}$
e^{at}	$\frac{1}{s-a}$	$e^{at} \sin \omega t$	$\frac{\omega}{(s-a)^2 + \omega^2}$
te^{-at}	$\frac{1}{(s+a)^2}$	$t^{n-1} e^{-at}$	$\frac{(n-1)!}{(s+a)^n}$

$$\text{ST}_x(p) = Px$$

$$\text{ST}_x(\perp) = \perp$$

$$\text{ST}_x(\neg\varphi) = \neg \text{ST}_x(\varphi)$$

$$\text{ST}_x(\varphi \wedge \psi) = \text{ST}_x(\varphi) \wedge \text{ST}_x(\psi)$$

$$\text{ST}_x(\langle m \rangle \varphi) = \exists y (R^m xy \wedge \text{ST}_y(\varphi))$$

$$\text{ST}_x([m] \varphi) = \forall y (R^m xy \rightarrow \text{ST}_y(\varphi))$$



Cyber Ontology Foundry

Organization designed to facilitate coordinated development of cyber ontologies.

1 follower

<https://ncor-network.org/docs/rese...>

info@ncornetwork.org

README.md



Cyber Ontology Foundry

The Cyber Ontology Foundry is a community-governed initiative to develop, maintain, and coordinate a suite of open, interoperable, formally grounded ontologies for the cyber domain.

Its purpose is to support shared understanding, data integration, automated reasoning, threat analysis, cyber mission planning, incident response, compliance, vulnerability management, and AI-enabled cyber operations across government, industry, academia, and standards communities.

Why the Cyber Ontology Foundry Exists

The cyber domain contains many important standards, schemas, taxonomies, and knowledge resources, including STIX/TAXII, ATT&CK, D3FEND, CAPEC, CVE, CWE, CPE, NIST frameworks, OSCAL, and related resources. These resources are valuable, but they were not all designed to share a common formal semantic foundation.

The Cyber Ontology Foundry provides a complementary semantic layer that helps clarify what these resources are about, how their entities relate, and how cyber data can be integrated, validated, queried, and reasoned over across systems.

The Foundry does not aim to replace existing cyber standards. It aims to make them more semantically interoperable.

<https://github.com/Cyber-Ontology-Foundry>

ry

2. Contents

Use this page to find the right contribution path.

- [3. Choose the right contribution path](#)
- [4. Basic rules for all contributions](#)
- [5. Submit an ontology for Foundry review](#)
 - [5.1 Possible review outcomes](#)
 - [5.2 What to include with an ontology submission](#)
 - [5.3 Minimum expectations for submitted ontologies](#)
- [6. Contribute to an existing Foundry ontology](#)
 - [6.1 Start with an issue](#)
 - [6.2 Request a new term](#)
 - [6.3 Improve a definition](#)
 - [6.4 Propose a new relation](#)
 - [6.5 Propose a mapping to an external standard](#)
 - [6.6 Contribute a design pattern](#)
 - [6.7 Add a use case or competency question](#)
 - [6.8 Report an ontology bug](#)
 - [6.9 Improve documentation](#)
- [7. Pull request expectations](#)
- [8. How review works](#)
- [9. Versioning, releases, and deprecation](#)
- [10. Security and sensitive content](#)
- [11. Licensing and intellectual property](#)
- [12. Communication norms](#)
- [13. Working group and Steering Committee review](#)

Contributing to the Cyber Ontology Foundry

Thank you for your interest in contributing to the Cyber Ontology Foundry.

The Cyber Ontology Foundry exists to develop, maintain, and coordinate open, interoperable, formally grounded ontologies for the cyber domain. Contributions should help make cyber data, standards, systems, observations, controls, vulnerabilities, incidents, threats, evidence, and claims more explicit, comparable, reusable, and machine-interpretable.

The Foundry does **not** aim to replace existing cyber standards. It aims to provide a formal semantic layer that helps align, clarify, validate, and reason across them.

This document explains how to:

- submit an ontology for Foundry review;
- contribute to an existing Foundry ontology;
- request new terms, definitions, relations, mappings, design patterns, use cases, and documentation changes;
- understand how Foundry review works;
- avoid security, sensitivity, licensing, and governance problems.

COF Steering Committee



John Beverley

Assistant Professor, *University at Buffalo*
President, National Center for Ontological
Research
Associate Director, *Institute of Artificial
Intelligence and Data Science*



Peter Kaloroumakis

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Ryan Hohimer

Hohimer Intelligence Strategies LLC

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The **Cyber Ontology Foundry** is a community-governed initiative to develop, maintain, and coordinate open, interoperable, formally grounded ontologies for the cyber domain.

The Foundry supports shared understanding, data integration, automated reasoning, threat analysis, cyber mission planning, incident response, compliance, vulnerability management, and AI-enabled cyber operations across government, industry, academia, and standards communities.

The Cyber Ontology Foundry supports NCOR's broader mission to advance ontology engineering best practices for trustworthy, interoperable data and AI systems.

Get involved

Interested in updates, working groups, ontology review, or contributing to the Cyber Ontology Foundry?

Sign up for the Cyber Ontology Foundry

You can also:

1. Visit the [Cyber Ontology Foundry GitHub organization](#).
2. Read the contribution guidelines.
3. Open an issue for a term, mapping, ontology submission, use case, or design pattern.
4. Join or propose a working group.
5. Participate in review and discussion.

For general NCOR information, visit the [NCOR Network homepage](#).

<https://ncor-network.org/docs/research/cyber-ontology-foundry>

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